

PROFESSIONAL SUMMARY Innovative Senior Electronics Hardware Design Engineer with over 10 years of R&D experience, specializing in taking complex hardware from concept to mass production. Expertise spans Industrial Electronics, Lift Control, Robotics, Consumer Electronics, Automotive, Defence Systems, and Biomedical fields. Adept at resolving critical signal integrity issues, optimizing power delivery networks (PDN), and ensuring strict adherence to military (MIL-STD) and industrial EMC/EMI standards. Proven track record of architecting complex hardware systems and leading engineering teams to deliver robust, high-performance electronic solutions, including requirements analysis, system partitioning, interface definition, risk mitigation, and cross-functional coordination.

CORE COMPETENCIES & TECHNICAL SKILLS

- Hardware Design:** Schematic Capture, PCB Layout (Rigid and Flex PCB up to 6 layers), Ethernet PHY up to 1 Gbps, USB 2.0/3.0 OTG, Mixed-Signal/Analog Design, Controlled Impedance Routing (Differential & Single-Ended), Length Matching and Skew Control, PDN & DFM Analysis
- Testing & Compliance:** Board bring-up and validation using custom STM32 firmware; Hardware debugging with oscilloscope, logic analyzer, etc, Schematic and PCB design EMC/EMI standards, Surge & Transient Protection, Signal Integrity Validation (Crosstalk, Eye Diagram), EMI pre-compliance test coordination, MIL-STD-810/461, EN 61000-4 series, IPC Standards, SIL-2.
- Microcontrollers & Tools:** ARM Cortex-M (STM32, TM4Cxx), FPGA (Efinix), Altium Designer, KiCad, LTSpice, Python (EDA Automation & Scripting), STM32CubeIDE, MATLAB.
- Power & Motor Control:** Buck/Boost Switching Regulators, PMIC optimization, Active/Passive Filters, High-precision ADCs, DC/BLDC/PMSM Motor Control Drivers.
- Communication Protocols:** RGMII, CAN Bus, USB, RS485, RS422, RS232, SPI, Quad SPI, I2C, UART, USART etc.

PROFESSIONAL EXPERIENCE

Istanbul Defence Systems @ Bilişim Vadisi, Kocaeli Turkey & @ Doha, Qatar ~ Senior Hardware Design Engineer April 2025 – Present

- Architect and develop high-reliability Gimbal Control Boards and Unmanned Aerial Vehicle (UAV) Mission Cards for advanced defense applications.
- Architected Complex Hardware Solutions and managed the end-to-end design process.
- Direct the hardware engineering team, overseeing workload distribution, mentoring junior engineers, and accelerating project delivery timelines.

GMT Industrial Electronic Ltd. @ Istanbul, Turkey ~ Senior Hardware Design Engineer July 2023 – March 2025

- Engineered Main Controller Boards and expansion modules for Programmable Logic Controllers (PLCs), integrating high-precision Delta Sigma ADC for RTD and Thermocouple interfaces.
- Optimized power management systems across PLC units to significantly reduce overall power consumption.
- Resolved field reliability issues by implementing robust surge protection circuitry and rigorous EMC/EMI filtering.

Acri Industrial Ltd. (Part of Ake Elevator & Escalator Ltd.) @ Antalya, Turkey ~ Senior Hardware Design Engineer April 2022 – June 2023

- Designed high-performance Lift BLDC Door Control Drivers and Lift Main Controllers.
- Developed an innovative LTE-based fault-tracking system enabling remote control and diagnostics.
- Designed custom Test Machine Main Boards to automate and streamline manufacturing processes.

Ake Elevator & Escalator Ltd. @ Antalya, Turkey ~ Senior Hardware Design Engineer Feb 2021 – April 2022

- Engineered top and bottom Lift Cabinet Control Boards, Lift Calling Boards, and Doppler Radar Sensor Boards for escalators.
- Designed Lift DC Door Control Drivers with integrated GSM capabilities for remote telemetry.

Desird Design R&D A.S. @ Antalya, Turkey ~ Mid-Level Hardware Design Engineer May 2018 – Feb 2021

- Developed linear motor drive systems for elevators and advanced BLDC door control drivers.
- Designed Railway Platform Separators utilizing SIL-2 certified safety relays to meet strict safety standards.

Desird Design R&D A.S. @ Antalya, Turkey ~ Junior Hardware Design Engineer Aug 2015 – Oct 2017

(Note: Completed 7-month mandatory military service during this period)

- Contributed to the complete hardware lifecycle for diverse projects, including biomedical devices (Lengthening Nails), IoT consumer electronics, energy telemetry systems, Motor Drivers etc.

Turkcell Communication A.S. @ Antalya, Turkey ~ Electrical and Electronic Engineering Intern July 2013 – Sept 2013

- Analyzed GSM signals and assisted in the strategic location planning of GSM Base Stations.

SELECTED PROJECTS

- Industrial BLDC Motor Control Board (24 VDC):** Designed a robust 4-layer PCB featuring CAN Bus, RS485, and absolute encoder interfaces. Integrated discrete MOSFET drivers (IR21xx) and differential op-amps for back-EMF current sensing. Developed custom STM32 firmware to verify hardware functionality and ensure system stability under load.

EDUCATION & CERTIFICATIONS

- Bachelor of Science in Electrical & Electronics Engineering**, Akdeniz University - Antalya, Turkey (Oct 2016)
- Certifications:** TUV Nord Safety Integrity Level for Railway System Electronics (2019) | English B2 Upper-Intermediate (2023)

HOBBIES

- Latin Dance** (Salsa & Bachata)